



AthenaStudies

Logic & Sets Day 2

Crash course | Mara Spadon | 2024

Planning

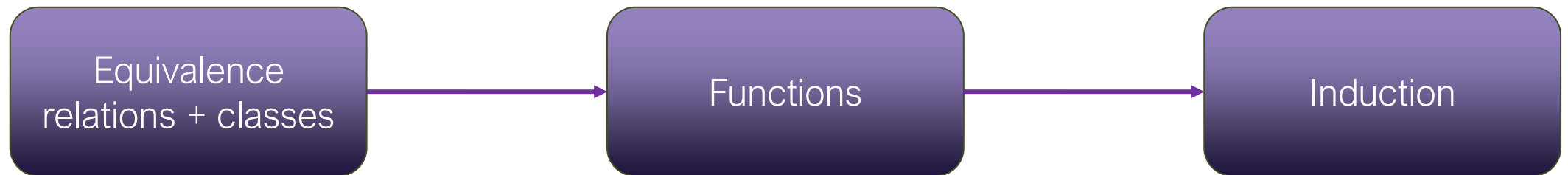
10-15 min break
every 45 min!

Wednesday	Logic week 5 + 6 + 7
Thursday	Sets week 5: ○ Equivalence relations + classes Sets week 6: ○ Functions Sets week 7: ○ Induction

WhatsApp



Planning Day 2



Equivalence relations / classes

Equivalence classes

Definition:

Equivalence relation is:

- Reflexive
- Transitive
- Symmetric

to check if R is
an equivalence
relation: check all 3! 

Let R be a relation on a set A .

① R is reflexive if

$$\forall x \in A, (x, x) \in R.$$



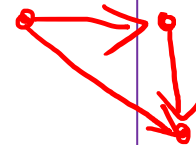
② R is symmetric if

$$\forall x, y \in A, (x, y) \in R \rightarrow (y, x) \in R.$$



③ R is transitive if

$$\forall x, y, z \in A, \\ (x, y) \in R \text{ and } (y, z) \in R \rightarrow (x, z) \in R.$$



all 3 have
to be true.
If one does
not hold
 $\Rightarrow$ not eq. rel.

Equivalence classes

Definition:

Equivalence relation is:

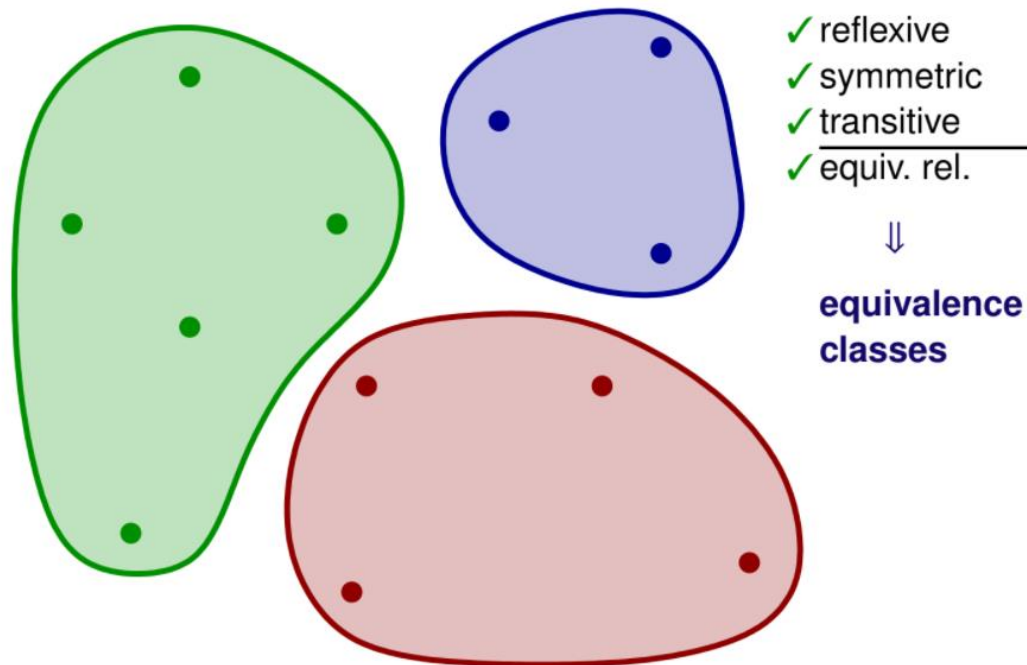
- Reflexive
- Transitive
- Symmetric

$\equiv$ is an equivalence relation!

- Can be used to construct equivalence classes
- Within an equivalence class of some relation R , all elements are related by this same relation R

Equivalence classes

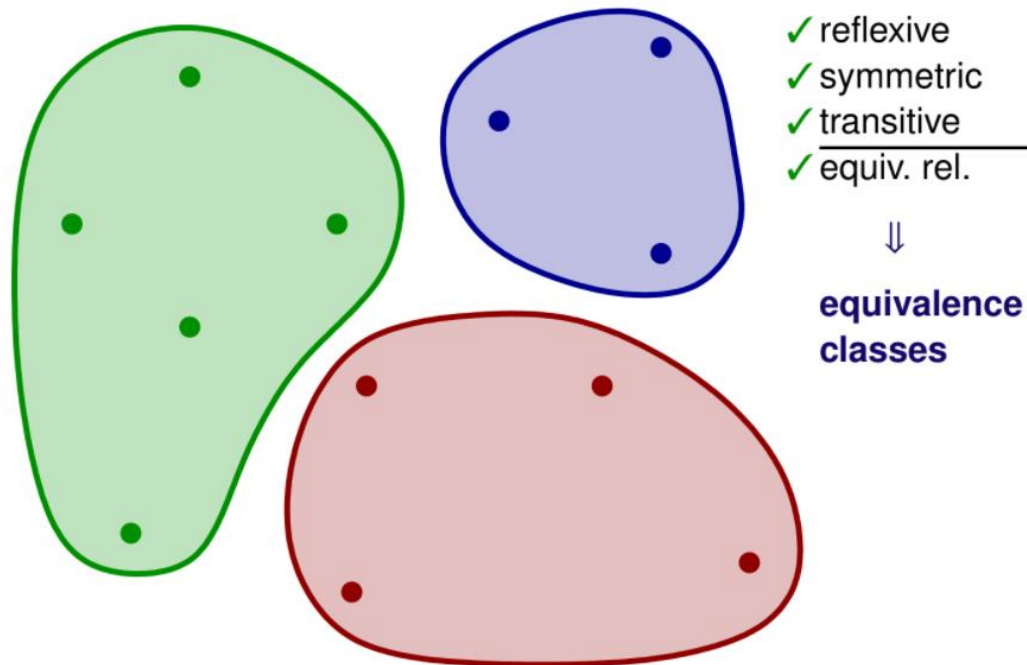
Equivalence class in sets:



The equivalence class of an element $a \in A$ is the set
 $[a] := \{x \in A : x \equiv a\}$

Equivalence classes

Equivalence class in sets:



The equivalence class of an element $a \in A$ is the set
 $[a] := \{x \in A : x \equiv a\}$

The equivalence classes of A are
a **partition** of A !
(they divide A in parts)

no overlap

Equivalence classes

The equivalence class of an element $a \in A$ is the set
 $[a] := \{x \in A : x \equiv a\}$

A **complete system of representatives** for $\equiv$ in A is a set $S \subseteq A$ that contains *precisely one* element from each equivalence class

last slide $\Rightarrow$ one red
 one green
 one blue

Equivalence classes: Example

Exam Question: Final 2021

1. Equivalence relations (7+8 points)

Suppose that in the set $A := \{0, 1, 2, 3, 4, 5\} \times \{0, 1, 2, 3, 4, 5\}$ we define the following relation

$$\langle a, b \rangle R \langle c, d \rangle :\Leftrightarrow a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3},$$

where $x \equiv y \pmod{m} :\Leftrightarrow y - x$ is a multiple of m .

- b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

Each equivalence class is made up of pairs $\langle a, b \rangle$ and $\langle c, d \rangle$ such that $\langle a, b \rangle R \langle c, d \rangle$

So, for every combination of pairs, we write down which other pairs it is equal to!

Equivalence classes: Example

Exam Question: Final 2021 $\langle 0,0 \rangle, \langle 0,1 \rangle, \langle 0,2 \rangle \dots \langle 5,5 \rangle$

1. Equivalence relations (7+8 points)

Suppose that in the set $A := \{0, 1, 2, 3, 4, 5\} \times \{0, 1, 2, 3, 4, 5\}$ we define the following relation

$$\langle a, b \rangle R \langle c, d \rangle \Leftrightarrow a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3},$$

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So, for every combination of pairs, we write down which other pairs it is equal to!

$\langle 0,0 \rangle R \langle 0,1 \rangle ?$

$$\langle 0,0 \rangle R \langle 0,1 \rangle$$

$$\begin{array}{ll} 0 - 0 = 0 & , \text{ mod } 2? \text{ yes} \\ 1 - 0 = 1 & , \text{ mod } 3? \text{ no} \end{array}$$

Equivalence classes: Example

Exam Question: Final 2021

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$$\langle a, b \rangle R \langle c, d \rangle :\Leftrightarrow a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3},$$

where $x \equiv y \pmod{m} :\Leftrightarrow y - x$ is a multiple of m .

- b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

$\langle 0, 0 \rangle R \langle 0, 1 \rangle ? \longrightarrow$ No, $0 \not\equiv 1 \pmod{3}$

Each equivalence class is made up of pairs $\langle a, b \rangle$ and $\langle c, d \rangle$ such that $\langle a, b \rangle R \langle c, d \rangle$

So, for every combination of pairs, we write down which other pairs it is equal to!

Equivalence classes: Example

Exam Question: Final 2021

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$$\langle a, b \rangle R \langle c, d \rangle :\Leftrightarrow a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3}, \quad \%$$

where $x \equiv y \pmod{m} :\Leftrightarrow y - x$ is a multiple of m .

- b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

$$\langle 0, 0 \rangle R \langle 2, 0 \rangle R \langle 4, 0 \rangle R \langle 0, 3 \rangle R \langle 2, 3 \rangle R \langle 4, 3 \rangle$$

So, first equivalence class:

$$\{\langle 0, 0 \rangle, \langle 2, 0 \rangle, \langle 4, 0 \rangle, \langle 0, 3 \rangle, \langle 2, 3 \rangle, \langle 4, 3 \rangle\}$$

$$\langle 0, 0 \rangle R \langle 4, 3 \rangle$$

$$4 - 0 = 4, \text{ mod } 2? \text{ yes}$$

$$3 - 0 = 3, \text{ mod } 3? \text{ yes}$$

Each equivalence class is made up of pairs $\langle a, b \rangle$ and $\langle c, d \rangle$ such that $\langle a, b \rangle R \langle c, d \rangle$

So, for every combination of pairs, we write down which other pairs it is equal to!

Equivalence classes: Example

Exam Question: Final 2021

1. Equivalence relations (7+8 points)

Suppose that in the set $A := \{0, 1, 2, 3, 4, 5\} \times \{0, 1, 2, 3, 4, 5\}$ we define the following relation

$$\langle a, b \rangle R \langle c, d \rangle \iff a \equiv c \pmod{2} \text{ and } b \equiv d \pmod{3},$$

where $x \equiv y \pmod{m} \iff y - x$ is a multiple of m .

- b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

- 1 $\{\langle 0, 0 \rangle, \langle 2, 0 \rangle, \langle 4, 0 \rangle, \langle 0, 3 \rangle, \langle 2, 3 \rangle, \langle 4, 3 \rangle\}$
- 2 $\{\langle 0, 1 \rangle, \langle 2, 1 \rangle, \langle 4, 1 \rangle, \langle 0, 4 \rangle, \langle 2, 4 \rangle, \langle 4, 4 \rangle\}$
- 3 $\{\langle 0, 2 \rangle, \langle 2, 2 \rangle, \langle 4, 2 \rangle, \langle 0, 5 \rangle, \langle 2, 5 \rangle, \langle 4, 5 \rangle\}$
- 4 $\{\langle 1, 0 \rangle, \langle 3, 0 \rangle, \langle 5, 0 \rangle, \langle 1, 3 \rangle, \langle 3, 3 \rangle, \langle 5, 3 \rangle\}$
- 5 $\{\langle 1, 1 \rangle, \langle 3, 1 \rangle, \langle 5, 1 \rangle, \langle 1, 4 \rangle, \langle 3, 4 \rangle, \langle 5, 4 \rangle\}$
- 6 $\{\langle 1, 2 \rangle, \langle 3, 2 \rangle, \langle 5, 2 \rangle, \langle 1, 5 \rangle, \langle 3, 5 \rangle, \langle 5, 5 \rangle\}$

Each equivalence class is made up of pairs $\langle a, b \rangle$ and $\langle c, d \rangle$ such that $\langle a, b \rangle R \langle c, d \rangle$

So, for every combination of pairs, we write down which other pairs it is equal to!

Equivalence classes: Example

Exam Question: Final 2021

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Suppose that in the set $A := \{0, 1, 2, 3, 4, 5\} \times \{0, 1, 2, 3, 4, 5\}$ we define the following relation

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where $x \equiv y \pmod{m} \Leftrightarrow y - x$ is a multiple of m .

b) Explicitly write down all the different equivalence classes of the equivalence relation R on A , and give a complete system of representatives.

- 1 $\{\langle 0, 0 \rangle, \langle 2, 0 \rangle, \langle 4, 0 \rangle, \langle 0, 3 \rangle, \langle 2, 3 \rangle, \langle 4, 3 \rangle\}$
- 2 $\{\langle 0, 1 \rangle, \langle 2, 1 \rangle, \langle 4, 1 \rangle, \langle 0, 4 \rangle, \langle 2, 4 \rangle, \langle 4, 4 \rangle\}$
- 3 $\{\langle 0, 2 \rangle, \langle 2, 2 \rangle, \langle 4, 2 \rangle, \langle 0, 5 \rangle, \langle 2, 5 \rangle, \langle 4, 5 \rangle\}$
- 4 $\{\langle 1, 0 \rangle, \langle 3, 0 \rangle, \langle 5, 0 \rangle, \langle 1, 3 \rangle, \langle 3, 3 \rangle, \langle 5, 3 \rangle\}$
- 5 $\{\langle 1, 1 \rangle, \langle 3, 1 \rangle, \langle 5, 1 \rangle, \langle 1, 4 \rangle, \langle 3, 4 \rangle, \langle 5, 4 \rangle\}$
- 6 $\{\langle 1, 2 \rangle, \langle 3, 2 \rangle, \langle 5, 2 \rangle, \langle 1, 5 \rangle, \langle 3, 5 \rangle, \langle 5, 5 \rangle\}$

System of representatives:

$$\{\langle 0, 0 \rangle^1, \langle 0, 1 \rangle^2, \langle 0, 2 \rangle^3, \langle 1, 0 \rangle^4, \langle 1, 1 \rangle^5, \langle 1, 2 \rangle^6\}$$

Each equivalence class is made up of pairs $\langle a, b \rangle$ and $\langle c, d \rangle$ such that $\langle a, b \rangle R \langle c, d \rangle$

So, for every combination of pairs, we write down which other pairs it is equal to!

System of representatives:

Pick one pair from each equivalence class!

Equivalence classes: Exercise!

Exam Question: Final 2022

6. Relations (3+5+4 points)

In the set $V = \{1, 2, 3, 4\}$ consider the relation R given by the set of pairs

$\{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle\}$.

a) Show that R is not an equivalence relation.

Reflexivity: $\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle$

Symmetry: $\langle 1, 2 \rangle \in R$ but $\langle 2, 1 \rangle \notin R$
So not sym.

$\Rightarrow$ not an equivalence relation

Definition:

Equivalence relation is:

- ✓ • Reflexive
- Transitive
- ✗ • Symmetric

✓ **Reflexivity**

for all elements x :
 $x R x$

Transitivity

for all elements x, y, z :
 $x R y \wedge y R z \rightarrow x R z$

Symmetry

✗ for all elements x, y :
 $x R y \rightarrow y R x$

Equivalence classes: Exercise!

Exam Question: Final 2022

6. Relations (3+5+4 points)

In the set $V = \{1, 2, 3, 4\}$ consider the relation R given by the set of pairs

$$\{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 4 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle\}.$$

a) Show that R is not an equivalence relation.

Solution: R is not an equivalence relation since R is not symmetric. For example, $\langle 1, 2 \rangle \in R$ but $\langle 2, 1 \rangle \notin R$.



Definition:

Equivalence relation is:

- Reflexive
- Transitive
- Symmetric

Reflexivity

for all elements x :

$$x R x$$

Transitivity

for all elements x, y, z :

$$x R y \wedge y R z \rightarrow x R z$$

Symmetry

for all elements x, y :

$$x R y \rightarrow y R x$$



Equivalence classes: Exercise!

Exam Question: Final 2019

5. Equivalence relations (6 + 9 points)

In this problem, we consider the relation R in the set $A := \{1, 2, 3, 6, 9\} \times \{1, 2, 3, 6, 9\}$ that is defined by the description

$$\langle a, b \rangle R \langle c, d \rangle \iff \frac{a}{b} = \frac{c}{d}$$

$$\frac{1}{1} = \frac{2}{2} \quad \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \dots\}$$

(a) Show that R is an equivalence relation.

Definition:

Equivalence relation is:

- Reflexive
- Transitive
- Symmetric

Reflexivity

for all elements x :
 $x R x$

Transitivity

for all elements x, y, z :
 $x R y \wedge y R z \rightarrow x R z$

Symmetry

for all elements x, y :
 $x R y \rightarrow y R x$

• Reflexivity: $\frac{a}{b} = \frac{a}{b}$, holds so $\langle a, b \rangle R \langle a, b \rangle$

• Transitivity: $\frac{a}{b} = \frac{c}{d}$, $\frac{c}{d} = \frac{e}{f}$, $\frac{a}{b} = \frac{e}{f}$ holds

• Symmetry: $\langle a, b \rangle R \langle c, d \rangle$ and $\langle c, d \rangle R \langle e, f \rangle$ impl. $\langle a, b \rangle R \langle e, f \rangle$

$\frac{a}{b} = \frac{c}{d}$ then also $\frac{c}{d} = \frac{a}{b}$ so $\langle a, b \rangle R \langle c, d \rangle$
 $\langle c, d \rangle R \langle a, b \rangle$

Equivalence classes: Exercise!

Exam Question: Final 2019

5. Equivalence relations (6 + 9 points)

In this problem, we consider the relation R in the set $A := \{1, 2, 3, 6, 9\} \times \{1, 2, 3, 6, 9\}$ that is defined by the description

$$\langle a, b \rangle R \langle c, d \rangle \iff \frac{a}{b} = \frac{c}{d}$$

(a) Show that R is an equivalence relation.

Solution: We have to show that R is reflexive, symmetric and transitive.

Reflexivity: $a/b = a/b$ hence $\langle a, b \rangle R \langle a, b \rangle$.

Symmetry: $a/b = c/d$ implies $c/d = a/b$, hence $\langle a, b \rangle R \langle c, d \rangle$ implies $\langle c, d \rangle R \langle a, b \rangle$.

Transitivity: $a/b = c/d$ and $c/d = e/f$ together imply $a/b = e/f$, hence we also have that $\langle a, b \rangle R \langle c, d \rangle$ and $\langle c, d \rangle R \langle e, f \rangle$ together imply $\langle a, b \rangle R \langle e, f \rangle$.

$$\frac{a}{b} = \frac{c}{d} \quad \frac{c}{d} = \frac{e}{f} \implies \frac{a}{b} = \frac{e}{f}$$


Definition:

Equivalence relation is:

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- Transitive
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Reflexivity

for all elements x :

$$x R x$$

Transitivity

for all elements x, y, z :

$$x R y \wedge y R z \rightarrow x R z$$

Symmetry

for all elements x, y :

$$x R y \rightarrow y R x$$

Equivalence classes: Exercise!

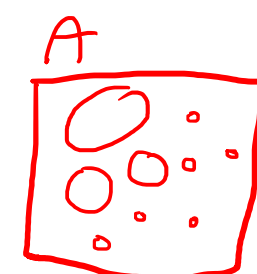
Exam Question: Final 2019

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$$\langle a, b \rangle R \langle c, d \rangle \iff \frac{a}{b} = \frac{c}{d}$$

25 pairs in total!



(a) Show that R is an equivalence relation.

(b) Explicitly write down all the different equivalence classes of the equivalence relation R in A , and give a complete system of representatives.

13 pairs

class could also include one single element

$$\begin{aligned}
 1 &= \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 6, 6 \rangle, \langle 9, 9 \rangle\} \\
 0.5 &= \{\langle 1, 2 \rangle, \langle 3, 6 \rangle\} \quad \{\langle 2, 3 \rangle, \langle 6, 9 \rangle\} \\
 2 &= \{\langle 2, 1 \rangle, \langle 6, 3 \rangle\} \quad \{\langle 3, 2 \rangle, \langle 9, 6 \rangle\} \\
 &\quad \{\langle 1, 3 \rangle, \langle 2, 6 \rangle, \langle 3, 9 \rangle\}, \{\langle 3, 1 \rangle, \langle 6, 2 \rangle, \langle 9, 3 \rangle\}
 \end{aligned}$$

Definition:

Equivalence relation is:

- Reflexive
- Transitive
- Symmetric

Reflexivity

for all elements x :
 $x R x$

Transitivity

for all elements x, y, z :
 $x R y \wedge y R z \rightarrow x R z$

Symmetry

for all elements x, y :
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Equivalence classes: Exercise!

Exam Question: Final 2019

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In this problem, we consider the relation R in the set $A := \{1, 2, 3, 6, 9\} \times \{1, 2, 3, 6, 9\}$ that is defined by the description

$$\langle a, b \rangle R \langle c, d \rangle \iff \frac{a}{b} = \frac{c}{d}$$

(a) Show that R is an equivalence relation.

(b) Explicitly write down all the different equivalence classes of the equivalence relation R in A , and give a complete system of representatives.

$$\begin{aligned} & \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 6, 6 \rangle, \langle 9, 9 \rangle\} \\ & \{\langle 1, 3 \rangle, \langle 2, 6 \rangle, \langle 3, 9 \rangle\} \quad \{\langle 3, 1 \rangle, \langle 6, 2 \rangle, \langle 9, 3 \rangle\} \\ & \{\langle 1, 2 \rangle, \langle 3, 6 \rangle\} \quad \{\langle 2, 1 \rangle, \langle 6, 3 \rangle\} \quad \{\langle 2, 3 \rangle, \langle 6, 9 \rangle\} \quad \{\langle 3, 2 \rangle, \langle 9, 6 \rangle\} \\ & \{\langle 1, 6 \rangle\} \quad \{\langle 1, 9 \rangle\} \quad \{\langle 2, 9 \rangle\} \quad \{\langle 6, 1 \rangle\} \quad \{\langle 9, 1 \rangle\} \quad \{\langle 9, 2 \rangle\} \end{aligned}$$

Hence, a complete system of representatives is for example

$$\begin{aligned} & \{\langle 1, 1 \rangle, \langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 1, 6 \rangle, \langle 1, 9 \rangle, \\ & \quad \langle 2, 1 \rangle, \langle 2, 3 \rangle, \langle 2, 9 \rangle, \langle 3, 1 \rangle, \langle 3, 2 \rangle, \langle 6, 1 \rangle, \langle 9, 1 \rangle, \langle 9, 2 \rangle\} \end{aligned}$$

NB: the equivalence classes consist of pairs $\langle a, b \rangle$ that represent the same rational number a/b .

Definition:

Equivalence relation is:

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- Transitive
- Symmetric

Reflexivity

for all elements x :

$$x R x$$

Transitivity

for all elements x, y, z :

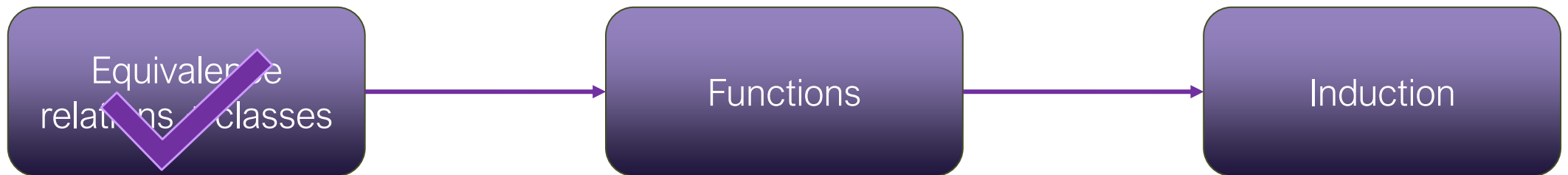
$$x R y \wedge y R z \rightarrow x R z$$

Symmetry

for all elements x, y :

$$x R y \rightarrow y R x$$

Planning Day 2



Break :)

Break time!

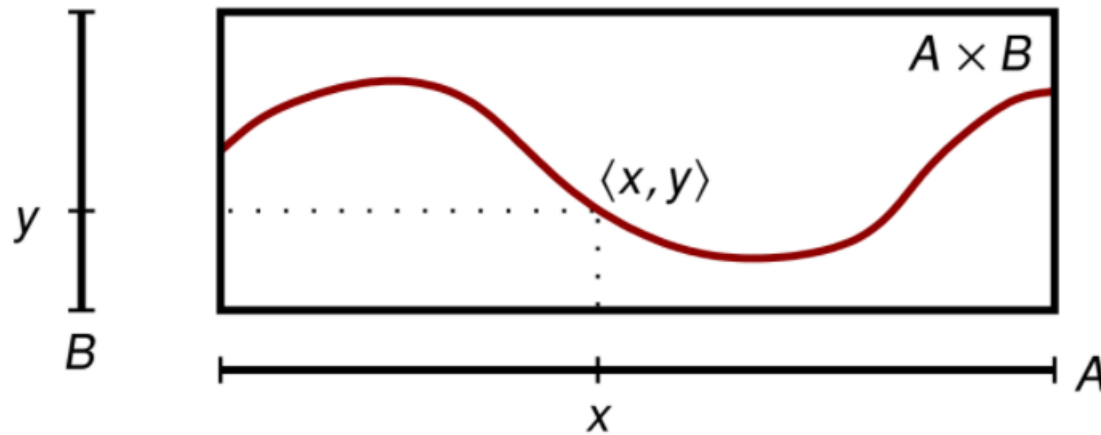
Meet back at 10:10

Functions

Functions

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$

$$f(x) = y$$



Functions

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$

domain A
codomain B

domain of definition $D_f := \{ \underline{x \in A} : f(x) = y \text{ for some } \underline{y \in B} \}$
range (or image) $R_f := \{ \underline{y \in B} : y = f(x) \text{ for some } \underline{x \in A} \}$

Functions

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$

domain A
codomain B

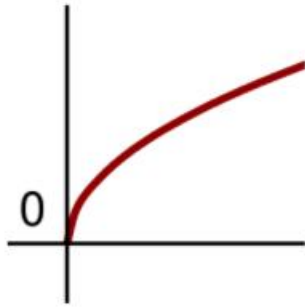
domain of definition $D_f := \{x \in A : f(x) = y \text{ for some } y \in B\}$
range (or image) $R_f := \{y \in B : y = f(x) \text{ for some } x \in A\}$

- ▶ if $D_f = A$ then f is called **total**
- ▶ if $R_f = B$ then f is called **surjective**

Functions: Example

Examples

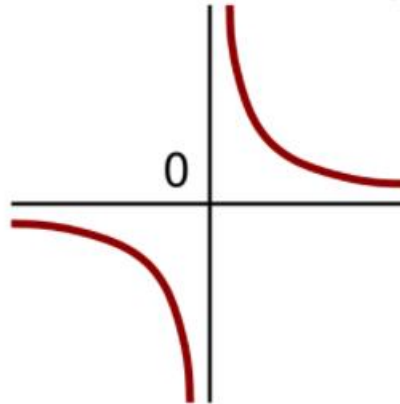
$f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \sqrt{x}$



$$D_f = \{x: x \geq 0\}$$

$$R_f = \{y: y \geq 0\}$$

$g: \mathbb{R} \rightarrow \mathbb{R}$ with $g(x) = \frac{1}{x}$



$$D_g = \mathbb{R} \setminus \{0\}$$

$$R_g = \mathbb{R} \setminus \{0\}$$

$D_f = \mathbb{R}$ total

$R_f = \mathbb{R}$ surjective

domain	A
codomain	B

domain of definition	$D_f := \{x \in A: f(x) = y \text{ for some } y \in B\}$
range (or image)	$R_f := \{y \in B: y = f(x) \text{ for some } x \in A\}$

Functions: Exercise!

Exam Question: resit 2020

7. Functions (6 + 4 points)

^A ^B ^A ^B ^A ^B

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) := \frac{1}{x^2 - 1} \quad mul_3(x) := 3x \quad g(x) := \sqrt{x - 1}$$

(a) What is the domain of definition of f , and what is its range?

What is the domain of definition of g , and what is its range?

$$f(x) = \frac{1}{x^2 - 1}$$

$(1)^2 = 1$
 $(-1)^2 = 1 \Rightarrow 0$ cannot divide by 0

$$D_f: \mathbb{R} \setminus \{1, -1\}$$

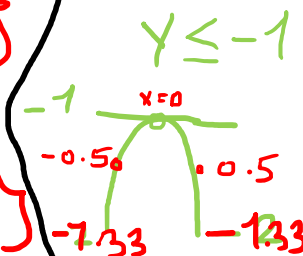
$$R_f: \{y \in \mathbb{R} \mid y > 0 \wedge y \leq -1\}$$

- For values > 1 , y approaches 0
- For values < -1 , y approaches 0

$$g(x) = \sqrt{x - 1}$$

$$D_g = \{x \in \mathbb{R} \mid x \geq 1\}$$

$$R_g = \{y \in \mathbb{R} \mid y \geq 0\}$$



$x=0 \rightarrow -1$
 $? \rightarrow -1$

$x=0.5 \rightarrow -1.333$

$x=-0.5 \rightarrow -2.333$

$app. 0 \leftarrow -1 \leftrightarrow 1 \rightarrow app. 0$

domain

$A: -0.5 \quad 0.5$

codomain

B

domain of definition

$D_f := \{x \in A : f(x) = y \text{ for some } y \in B\}$

range (or image)

$R_f := \{y \in B : y = f(x) \text{ for some } x \in A\}$

Functions: Exercise!

Exam Question: resit 2020

7. Functions (6 + 4 points)

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

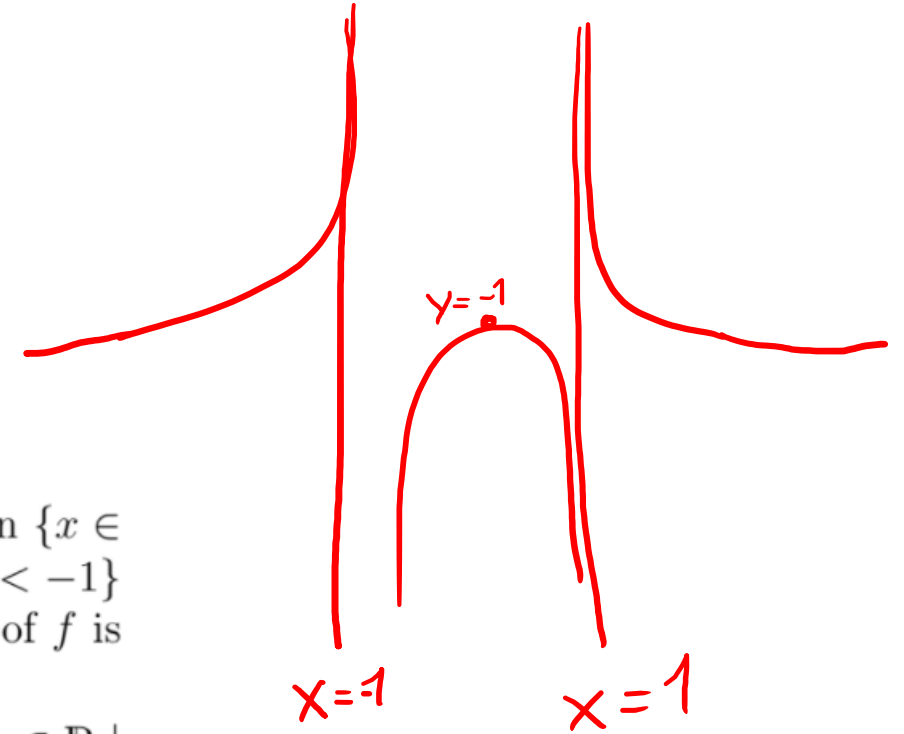
$$f(x) := \frac{1}{x^2 - 1} \quad mul_3(x) := 3x \quad g(x) := \sqrt{x - 1}$$

(a) What is the domain of definition of f , and what is its range?

What is the domain of definition of g , and what is its range?

Solution: The domain of definition of f is $\mathbb{R} \setminus \{-1, 1\}$. The range of f on $\{x \in \mathbb{R} \mid -1 < x < 1\}$ is $\{y \in \mathbb{R} \mid y \leq -1\}$ and the range of f on $\{x \in \mathbb{R} \mid x < -1\}$ as well as on $\{x \in \mathbb{R} \mid x > 1\}$ is $\{y \in \mathbb{R} \mid y > 0\}$. So overall the range of f is $\{y \in \mathbb{R} \mid y \leq -1 \vee y > 0\}$.

The domain of definition of g is $\{x \in \mathbb{R} \mid x \geq 1\}$ and the range of g is $\{y \in \mathbb{R} \mid y \geq 0\}$.



domain	A
codomain	B
domain of definition	$D_f := \{x \in A : f(x) = y \text{ for some } y \in B\}$
range (or image)	$R_f := \{y \in B : y = f(x) \text{ for some } x \in A\}$

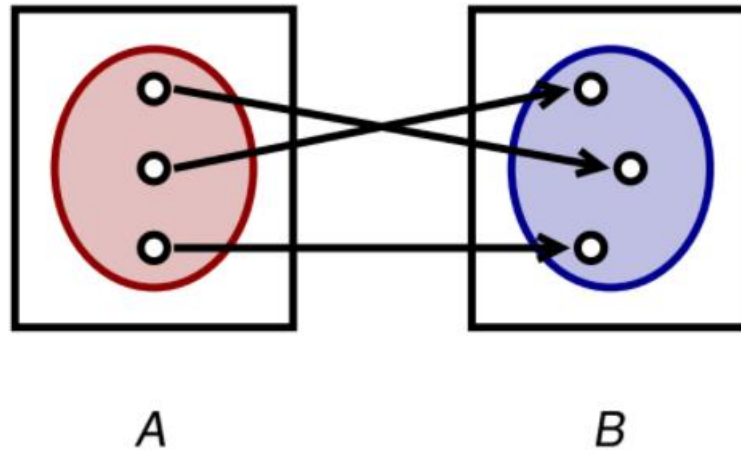
Functions

$f: A \rightarrow B$ is **injective** if

$$x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$$

or equivalently, if

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2$$



'every $x \in A$ maps to a
unique $b \in B$ '

exactly one

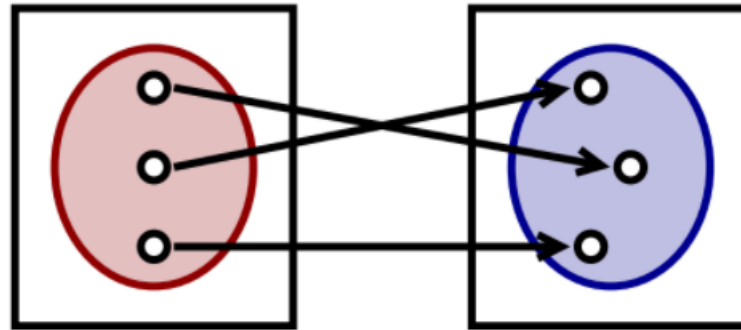
Functions

$f: A \rightarrow B$ is **injective** if

$$x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$$

or equivalently, if

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2$$



A

B

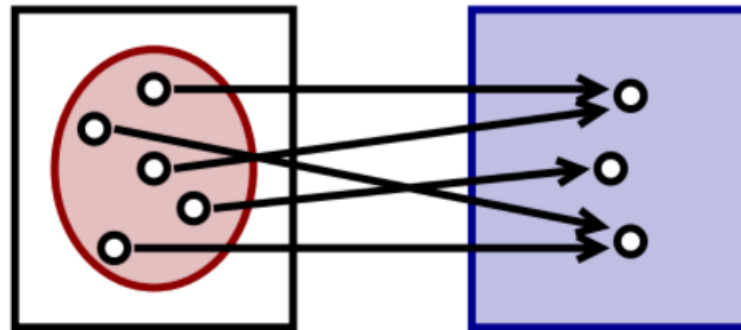
‘every $x \in A$ maps to a unique $b \in B$ ’

exactly one

$f: A \rightarrow B$ is **surjective** if

$$R_f = B$$

is Range =
codomain?



‘for every $b \in B$, there is an $x \in A$ that maps to b ’

at least one

“Range is equal to the codomain”

range (or image)

$$R_f := \{y \in B : y = f(x) \text{ for some } x \in A\}$$

$R = R \checkmark$

Functions: Exercise!

Exam Questions: resit 2021

7. Functions (5+6 points)

$A \rightarrow B$

We are given the functions $sqr : \mathbb{R} \rightarrow \mathbb{R}$, $mul_3 : \mathbb{R} \rightarrow \mathbb{R}$, $sub_2 : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$sqr(x) = x^2, \quad mul_3(x) = 3x, \quad sub_2(x) = x - 2.$$

(a) For each function, determine whether or not the function is injective.

$$\div 2 \quad \begin{cases} 2x = 2y \\ \Rightarrow x = y \end{cases}$$

$mul_3(x)?$

$$\begin{cases} 3x = 3y \\ \Rightarrow x = y \end{cases} \quad \checkmark$$

injective

$sub_2(x)?$

$$\begin{cases} x-2 = y-2 \\ \Rightarrow x = y \end{cases} \quad \checkmark$$

injective

Injective: 'every $x \in A$ maps to a unique $b \in B$ '

$$sqr(x) = x^2 \Rightarrow \text{not injective}$$

$$sqr(1) = sqr(-1)$$

$$x^2 = y^2 \quad x^2 = y^2$$

$$\sqrt{x} = \sqrt{y}$$

$$-2^2 = 2^2$$

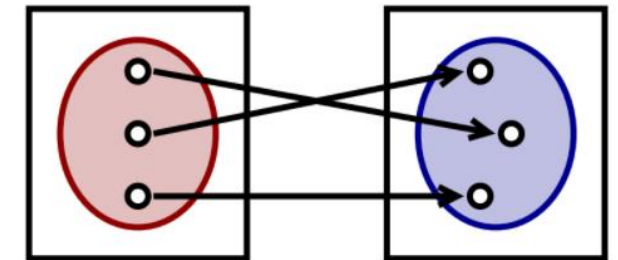
not the same

$f: A \rightarrow B$ is **injective** if

$$x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$$

or equivalently, if

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2$$



A

B

Functions: Exercise!

Exam Questions: resit 2021

7. Functions (5+6 points)

We are given the functions $sqr : \mathbb{R} \rightarrow \mathbb{R}$, $mul_3 : \mathbb{R} \rightarrow \mathbb{R}$, $sub_2 : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$sqr(x) = x^2, \quad mul_3(x) = 3x, \quad sub_2(x) = x - 2.$$

(a) For each function, determine whether or not the function is injective.

Solution: No, sqr is not injective, since e.g. $sqr(-1) = sqr(+1)$;

Yes, mul_3 is injective, since $3x = 3y$ gives $x = y$;

Yes, sub_2 is injective, since $x - 2 = y - 2$ gives $x = y$.

$$x_1^2 = x_2^2$$

$$x^2 = y^2$$

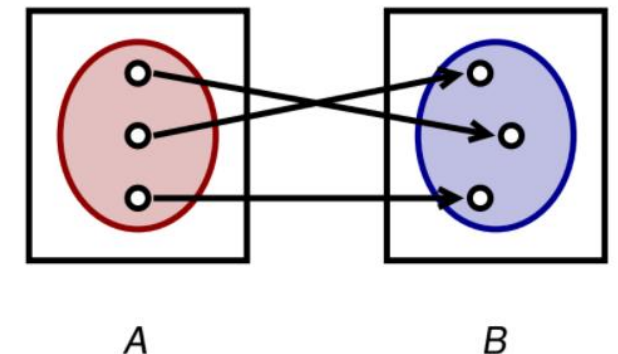
$f : A \rightarrow B$ is **injective** if

$$x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$$

or equivalently, if

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2$$

Injective: 'every $x \in A$ maps to a unique $b \in B$ '



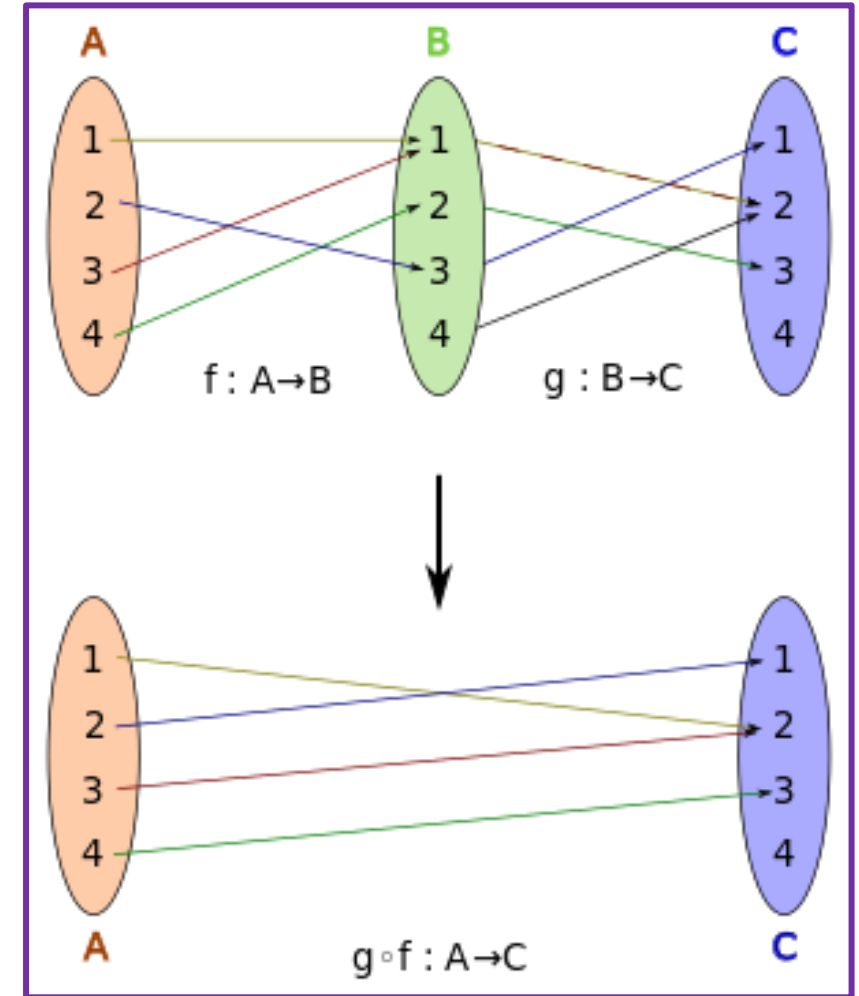
Functions

The **composition** $g \circ f$ of $f: A \rightarrow B$ and $g: B \rightarrow C$ satisfies

$$f(x) = y \wedge g(y) = z \quad \Rightarrow \quad (g \circ f)(x) = g(f(x)) = z$$

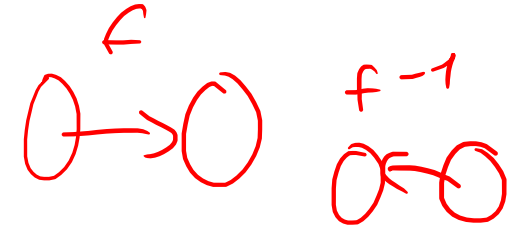
$$\downarrow$$

$$g(f(x)) = z$$



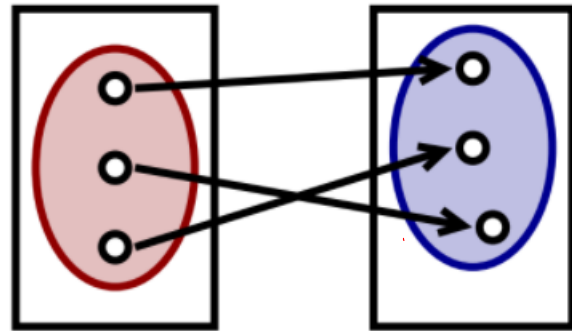
Functions

- ▶ $f: A \rightarrow B$ considered as a *relation* has an inverse
- ▶ this inverse *relation* is not necessarily a *function*



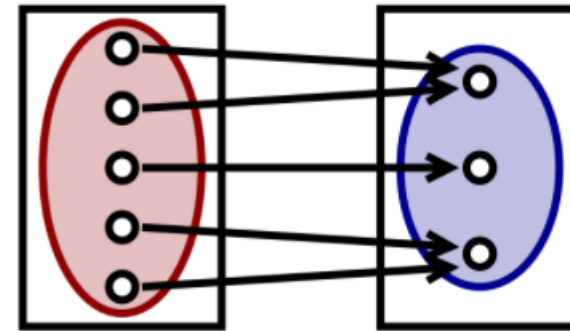
Theorem

The inverse of f is a function if and only if f is injective



injective: **one-one** or **1-1**

↳ inverse function



non-injective: **many-one**

↳ no inverse function

Functions

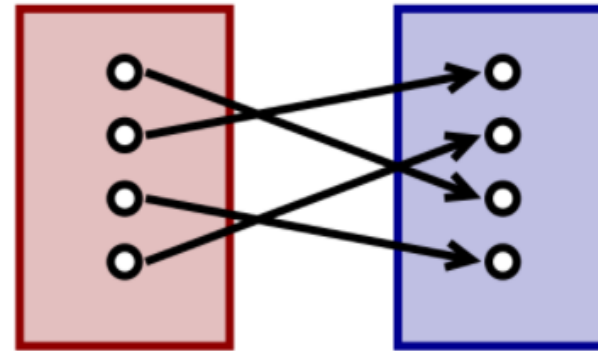
- ▶ a *total* injection $f: A \rightarrow B$ has an inverse function $f^{-1}: B \rightarrow A$
- ▶ this inverse function is not necessarily *total*

$f: A \rightarrow B$ is **bijective** if

- ▶ f is total
- ▶ f is injective
- ▶ f is surjective

at least one

Bijections have a bijective inverse



Functions: Example

Exam Question: resit 2020

7. Functions (6 + 4 points)

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) := \frac{1}{x^2 - 1} \quad mul_3(x) := 3x \quad g(x) := \sqrt{x - 1}$$

(b) We define $h = f \circ mul_3^{-1} \circ g$. Give an explicit formula for $h(x)$.

Functions: Example

Exam Question: resit 2020

7. Functions (6 + 4 points)

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

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(b) We define $h = f \circ mul_3^{-1} \circ g$. Give an explicit formula for $h(x)$.

$$f = \frac{1}{x^2 - 1}$$

$$m: y = 3x$$

$$m^{-1}: x = 3y \quad \div 3 \quad \text{Swap } x \text{ and } y$$

$$y = \frac{x}{3}$$

$$\frac{1}{3} \cdot x = \frac{x}{3}$$

Functions: Example

Exam Question: resit 2020

7. Functions (6 + 4 points)

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

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$$f = \frac{1}{\textcolor{red}{x}^2 - 1}$$

$$f \circ mul_3^{-1} = \frac{1}{(\frac{\textcolor{red}{x}}{3})^2 - 1}$$

Functions: Example

Exam Question: resit 2020

7. Functions (6 + 4 points)

We are given the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $mul_3: \mathbb{R} \rightarrow \mathbb{R}$, and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) := \frac{1}{x^2 - 1} \quad mul_3(x) := 3x \quad g(x) := \sqrt{x - 1}$$

(b) We define $h = f \circ mul_3^{-1} \circ g$. Give an explicit formula for $h(x)$.

$$f = \frac{1}{\textcolor{red}{x}^2 - 1}$$

$$f \circ mul_3^{-1} = \frac{1}{(\textcolor{red}{x}/3)^2 - 1}$$

$$f \circ mul_3^{-1} \circ g = \frac{1}{\left(\frac{\sqrt{x-1}}{3}\right)^2 - 1}$$

$f \circ g = g$ after f

Functions: Exercise!

Exam Question: final 2021

2. Functions and cardinality (5+5+5 points)

We are given the functions $\exp : \mathbb{R} \rightarrow \mathbb{R}$, $\text{add} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{root} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\exp(x) = 2^x, \text{add}(x) = x + 1, \text{ and } \text{root}(x) = \sqrt{x}.$$

- b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by $f = \exp \circ \text{root} \circ \text{add}^{-1}$.

$$\begin{aligned} \text{add}^{-1} : \quad & y = x + 1 \quad \text{swap } x \text{ \& } y \\ & x = y + 1 \quad \text{swap } x \text{ \& } y \\ & y = x - 1 \quad \text{swap } x \text{ \& } y \end{aligned}$$

$$f = 2^{\sqrt{x-1}}$$

Functions: Exercise!

Exam Question: final 2021

2. Functions and cardinality (5+5+5 points)

We are given the functions $\exp : \mathbb{R} \rightarrow \mathbb{R}$, $\text{add} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{root} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\exp(x) = 2^x, \text{add}(x) = x + 1, \text{ and } \text{root}(x) = \sqrt{x}.$$

- b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by $f = \exp \circ \text{root} \circ \text{add}^{-1}$.

Solution: $f(x) = (\exp \circ \text{root} \circ \text{add}^{-1})(x) = (\exp \circ \text{root})(x - 1) = \exp(\sqrt{x - 1}) = 2^{\sqrt{x - 1}}.$

Functions: Exercise!

Exam Question: final 2021

2. Functions and cardinality (5+5+5 points)

We are given the functions $\exp : \mathbb{R} \rightarrow \mathbb{R}$, $\text{add} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{root} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\exp(x) = 2^x, \text{add}(x) = x + 1, \text{and } \text{root}(x) = \sqrt{x}.$$

c) Express the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(x) = \sqrt{2^x} - 2$$

as a composition of $\exp$, add , root , and their inverses.

$$g(x) = \sqrt{2^x} - 2$$

$$\bullet 2^x$$

$$\bullet x + 1 \rightarrow \text{add}^{-1} = x - 1$$

$$\bullet \sqrt{x}$$

$$\begin{array}{ccc} 2^x & \exp & \uparrow \\ \sqrt{2^x} & \text{root} & \uparrow \\ \sqrt{2^x} - 1 & \text{add}^{-1} & \uparrow \end{array}$$

$$\begin{array}{c} \sqrt{2^x} - 2 \text{ add}^{-1} \\ \text{Flip around!} \\ = \text{add}^{-1} \circ \text{add}^{-1} \circ \text{root} \circ \exp \\ \sqrt{2^x} - 2 \end{array}$$

Functions: Exercise!

Exam Question: final 2021

2. Functions and cardinality (5+5+5 points)

We are given the functions $exp : \mathbb{R} \rightarrow \mathbb{R}$, $add : \mathbb{R} \rightarrow \mathbb{R}$, $root : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$exp(x) = 2^x, \quad add(x) = x + 1, \quad \text{and} \quad root(x) = \sqrt{x}.$$

c) Express the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\underline{g(x) = \sqrt{2^x} - 2}$$

as a composition of exp , add , $root$, and their inverses.

$$\begin{aligned} \text{Solution: } \sqrt{2^x} - 2 &= add^{-1}(\sqrt{2^x} - 1) = (add^{-1} \circ add^{-1})(\sqrt{2^x}) \\ &= (add^{-1} \circ add^{-1} \circ root)(2^x) = \underline{(add^{-1} \circ add^{-1} \circ root \circ exp)(x)}. \end{aligned}$$

Planning Day 2



Break :)

Break time!

Meet back at

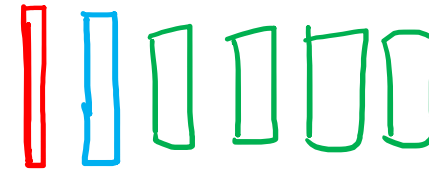


Induction

Principle of Induction

smallest value

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$



Domino effect!

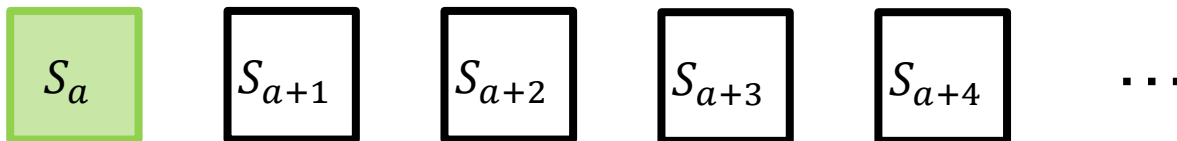


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

1. Show S_a

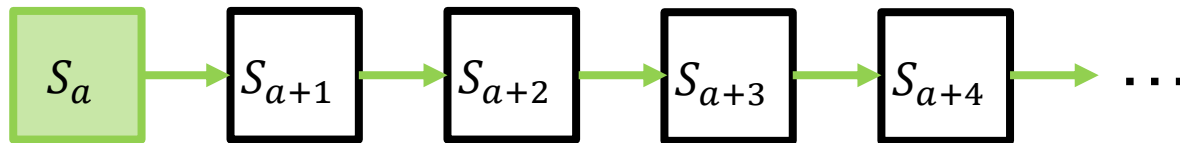


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

2. Show the implication

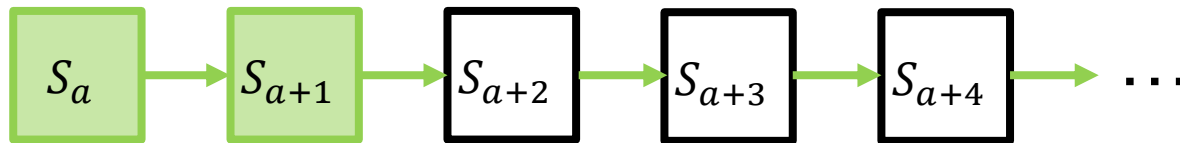


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

Proof complete 😊

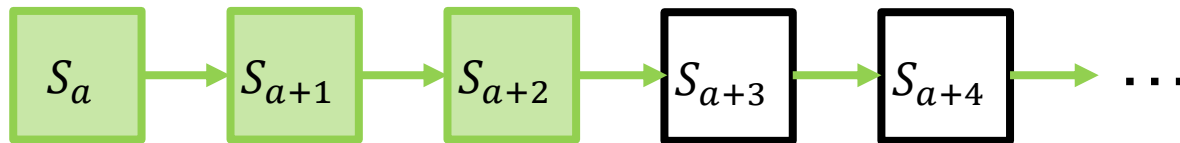


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

Proof complete 😊

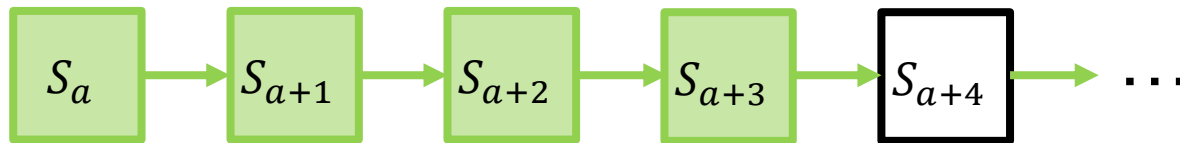


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

Proof complete 😊

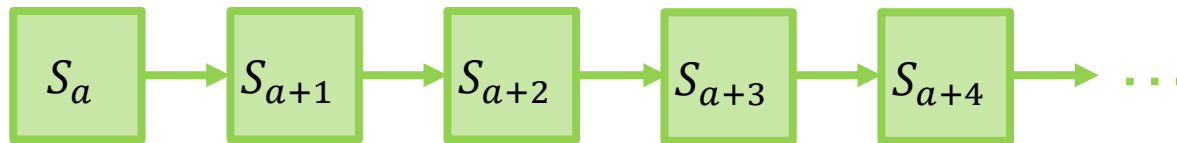


Principle of Induction

- IDEA:
1. Show statement holds for $n = a$
 2. Show that if it holds for m , then the statement also holds for $m + 1$

Domino effect!

Proof complete 😊



Steps

1. Prove S_a is true 1 Base case
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$ Induction step

Steps

1. Prove S_a is true

Base case

2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$

Induction step

Steps

1. Prove S_a is true



Base case

2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$

Induction step

- Let $m \geq a$ be arbitrary

Steps

1. Prove S_a is true

Base case

2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$

Induction step

- Let $m \geq a$ be arbitrary

Steps

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$

Base case

Induction step

Induction hypothesis (IH)

formula/rule/
theorem that
we use later
on! 

Steps

1. Prove S_a is true

Base case

2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$

Induction step

- Let $m \geq a$ be arbitrary
- Assume S_n is true for $n = m$
- Use IH to show S_{m+1}

Induction hypothesis (IH)

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k$ = $2^{n+1} - 1$

Steps:

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Base case: $n = ?$

Steps:

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Base case: $n = 0$

Steps:

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = \underline{2^{n+1} - 1}$

Base case: $n = 0$

- $\sum_{k=0}^{n=0} 2^k = 2^0 = 1$

Steps:

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Base case: $n = 0$

- $\sum_{k=0}^{n=0} 2^k = 2^0 = 1$
- $2^{n+1} - 1 = 2^{0+1} - 1 = 2 - 1 = 1$

Steps:

1. Prove S_a is true
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Base case: $n = 0$

- $\sum_{k=0}^0 2^k = 2^0 = 1$
- $2^{0+1} - 1 = 2^{0+1} - 1 = 2 - 1 = 1$

Conclusion: statement is true for $n = 0$

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

$$n \geq 0$$

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

$$\sum_{k=0}^2 2^k = 2^0 + 2^1 + 2^2$$

Steps:

1. ✓ Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$

- ✓ • Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH) $\rightarrow$ use this later on

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

✓ ✓
 $\boxed{a} \boxed{m} \underbrace{\boxed{0} \boxed{1} \boxed{2} \boxed{3}}_{m+1}$

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH)

⋮

So, IH implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH)

- $\sum_{k=0}^{m+1} 2^k =$

So, IH implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
 - Use IH to show S_{m+1}

Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH)

$$- \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1}$$

So, IH implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$

Steps:

1. Prove S_a is true 
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
 - Assume S_n is true for $n = m$ (IH)
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Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$

Induction step:

Let $m \geq 0$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH)

$$- \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1} \stackrel{\text{IH}}{=} 2^{m+1} - 1 + 2^{m+1}$$

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$$- \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1} = 2^{m+1} - 1 + 2^{m+1}$$

$$- 2^{\underline{(m+1)+1}} - 1 =$$

So, IH implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$

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$$- \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1} = 2^{m+1} - 1 + 2^{m+1}$$

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So, IH implies $\sum_{k=0}^{m+1} 2^k = 2^{(m+1)+1} - 1$

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Example

Claim: For all $n \in \mathbb{N}$, $\sum_{k=0}^n 2^k = 2^{n+1} - 1$
 $n=m+1$ $n=m+1$

Induction step:

Let $m \geq \underline{0}$.

Assume $\sum_{k=0}^m 2^k = 2^{m+1} - 1$ (IH)

$$- \sum_{k=0}^{m+1} 2^k = \sum_{k=0}^m 2^k + 2^{m+1} = 2^{m+1} - 1 + 2^{m+1}$$

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1. Prove S_a is true 
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 - Let $m \geq \underline{a}$ be arbitrary
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3. • Use IH to show S_{m+1}

Exercise!

Exam Question: resit 2020

8. Induction (12.5 points)

Prove using mathematical induction that the following statement is true for all integer numbers $n \geq 1$:

$$\sum_{k=1}^n (k-1)^3 = \frac{((n-1)n)^2}{4}$$

① Base case $n=1$

$$(1-1)^3 = 0$$

$$\frac{((1-1) \cdot 1)^2}{4} = \frac{(0 \cdot 1)^2}{4} = 0$$

$\Rightarrow$ Base case holds! ∇

② Let $m \geq 1$. Assume true for $n=m$

$$\sum_{u=1}^m (u-1)^3 = \frac{((m-1)m)^2}{4} \quad \text{IH}$$

③ Prove true for $m+1$:

$$\text{LHS: } \sum_{u=1}^{m+1} (u-1)^3 = \sum_{u=1}^m (u-1)^3 + ((m+1)-1)^3 = \dots$$

$$\text{RHS: } \frac{((m+1-1)(m+1))^2}{4} = \frac{(m(m+1))^2}{4}$$

Steps:

1. Prove S_a is true $\checkmark$
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary $\checkmark$
 - Assume S_n is true for $n = m$ $\checkmark$ (IH)
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Exercise!

Exam Question: resit 2020

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$n+1$

$n+1$

IH

RHS:

$$\frac{((n+1-1)(n+1))^2}{4} = \frac{(n(n+1))^2}{4}$$

Solution: In the base case $n = 1$, both the left- and the right-hand side of this equation are 0.

Now consider the inductive case:

LHS

$$\begin{aligned} \sum_{k=1}^{n+1} (k-1)^3 &= \sum_{k=1}^n (k-1)^3 + n^3 = \frac{((n-1)n)^2}{4} + n^3 \quad (\text{induction hypothesis}) \\ &= \frac{n^4 - 2n^3 + n^2}{4} + n^3 = \frac{n^4 + 2n^3 + n^2}{4} = \frac{(n(n+1))^2}{4} \end{aligned}$$

$$\begin{aligned} \sum_{k=1}^{n+1} (k-1)^3 &= \sum_{k=1}^n (k-1)^3 + ((n+1)-1)^3 \\ &= \sum_{k=1}^n (k-1)^3 + n^3 \end{aligned}$$

Steps:

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Exam Question: resit 2020

3. Induction (5 + 10 points)

We claim that the following statement is true for all integer numbers $n \geq 0$:

$$\sum_{k=0}^n 2(k+1) = (n+1)(n+2)$$

(recall that $\sum_{k=0}^n a_k$ is shorthand notation for $a_0 + a_1 + a_2 + \cdots + a_n$).

- (a) Verify by explicit calculation that the statement is true for $n = 0$, $n = 1$ and $n = 2$.

Exam Question: resit 2020

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- (a) Verify by explicit calculation that the statement is true for $n = 0$, $n = 1$ and $n = 2$.

Solution:

	$2(n+1)$	$\sum_{k=0}^n 2(k+1)$	$(n+1)(n+2)$
$n = 0$	$2 \cdot 1 = 2$	2	$1 \cdot 2 = 2$
$n = 1$	$2 \cdot 2 = 4$	$2 + 4 = 6$	$2 \cdot 3 = 6$
$n = 2$	$2 \cdot 3 = 6$	$2 + 4 + 6 = 12$	$3 \cdot 4 = 12$

So the statement is indeed true for $n = 0$, $n = 1$ and $n = 2$.

Exercise!

Exam Question: resit 2020

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Exercise!

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Exercise!

Exam Question: resit 2020

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 - Let $m \geq a$ be arbitrary
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Solution: We have already shown in part (a) that the statement is true for $n = 0$ (the base case). For the induction step, let $m \geq 0$ be arbitrary, and assume (as induction hypothesis) that

$$\sum_{k=0}^m 2(k+1) = (m+1)(m+2). \quad \text{IH}$$

Then, on the one hand,

$$\begin{aligned} \sum_{k=0}^{m+1} 2(k+1) &= \sum_{k=0}^m 2(k+1) + 2((m+1)+1) \\ &\stackrel{\text{IH}}{=} (m+1)(m+2) + 2(m+2) \\ &= ((m+1)+2)(m+2) \\ &= (m+2)(m+3). \end{aligned}$$

On the other hand,

$$((m+1)+1)((m+1)+2) = (m+2)(m+3).$$

Therefore, the induction hypothesis for $m \geq 0$ implies that

$$\sum_{k=0}^{m+1} 2(k+1) = ((m+1)+1)((m+1)+2).$$

Together with the base case, this proves that the statement in the claim is true for every integer $n \geq 0$.

Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax



Induction: Sequences

A total function $f: \mathbb{N} \rightarrow A$ defines the **sequence**

$$(f(n))_{n=0}^{\infty} = (f(0), f(1), f(2), \dots)$$

- ▶ $f(n)$ is called the **n -th term** of the sequence
- ▶ we also use the notation $(t_n)_{n=a}^{\infty} = (t_a, t_{a+1}, t_{a+2}, \dots)$

Induction: Sequences: Example

Claim the sequence $t_3 := 0, t_{n+1} := t_n + (n - 1)$ satisfies

$$t_n = \frac{1}{2}n(n - 3) \quad \text{for all } n \geq 3$$

Steps:

1. Prove S_a is true
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Proof – base case

- ▶ $t_3 = 0$
- ▶ $\frac{1}{2} \cdot 3 \cdot (3 - 3) = 0$

Hence $t_n = \frac{1}{2}n(n-3)$ is true for $n = 3$

Steps:

1. Prove S_a is true ✓
2. Prove $S_m \rightarrow S_{m+1}$ is true for any $m \geq a$
 - Let $m \geq a$ be arbitrary
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induction step

Let $m \geq 3$. Assume $t_m = \frac{1}{2}m(m - 3)$

IH

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Hence **IH** implies $t_{m+1} = \frac{1}{2}(m + 1)((m + 1) - 3)$



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Let $m \geq 3$. Assume $t_m = \frac{1}{2}m(m - 3)$

IH

► $t_{m+1} = t_m + (m - 1)$

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induction step

Let $m \geq 3$. Assume

$$t_m = \frac{1}{2}m(m - 3)$$

IH

$$\triangleright t_{m+1} = t_m + (m - 1) \stackrel{\text{IH}}{=} \frac{1}{2}m(m - 3) + (m - 1)$$

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induction step

Let $m \geq 3$. Assume $t_m = \frac{1}{2}m(m - 3)$

$$\begin{aligned} \blacktriangleright \quad t_{m+1} &= t_m + (m - 1) \stackrel{\text{IH}}{=} \frac{1}{2}m(m - 3) + (m - 1) \\ &= \frac{1}{2}m^2 - \frac{3}{2}m + m - 1 = \frac{1}{2}m^2 - \frac{1}{2}m - 1 \end{aligned}$$

Hence **IH** implies $t_{m+1} = \frac{1}{2}(m + 1)((m + 1) - 3)$

IH



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IH

$$\begin{aligned} \blacktriangleright t_{m+1} &= t_m + (m-1) \stackrel{\text{IH}}{=} \frac{1}{2}m(m-3) + (m-1) \\ &= \frac{1}{2}m^2 - \frac{3}{2}m + m - 1 = \frac{1}{2}m^2 - \frac{1}{2}m - 1 \\ \blacktriangleright \frac{1}{2}(m+1)((m+1)-3) &= \frac{1}{2}(m+1)(m-2) \end{aligned}$$

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IH

$$\begin{aligned} \blacktriangleright t_{m+1} &= t_m + (m - 1) \stackrel{\text{IH}}{=} \frac{1}{2}m(m - 3) + (m - 1) \\ &= \frac{1}{2}m^2 - \frac{3}{2}m + m - 1 = \frac{1}{2}m^2 - \frac{1}{2}m - 1 \\ \blacktriangleright \frac{1}{2}(m + 1)((m + 1) - 3) &= \frac{1}{2}(m + 1)(m - 2) \\ &= \frac{1}{2}(m^2 + m - 2m - 2) = \frac{1}{2}m^2 - \frac{1}{2}m - 1 \end{aligned}$$

Hence **IH** implies $t_{m+1} = \frac{1}{2}(m + 1)((m + 1) - 3)$



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Induction: Sequences: Exercise!

Exam Question: resit 2021

8. Induction and recursion (3+8 points)

Consider the sequence $(t_n)_{n \in \mathbb{N}}$ of numbers defined recursively by

$$t_0 := \frac{1}{2}, \quad t_{n+1} := t_n + 3^n + \frac{1}{2}.$$

We claim that the following statement is true for all natural numbers n :

$$t_n = \frac{1}{2}(3^n + n).$$

(a) Verify by explicit computation that the claim is true for $n = 0$, $n = 1$ and $n = 2$.

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$$t_n = \frac{1}{2}(3^n + n).$$

(a) Verify by explicit computation that the claim is true for $n = 0$, $n = 1$ and $n = 2$.

Solution: $t_0 = \frac{1}{2} = \frac{1}{2}(3^0 + 0);$
 $t_1 = \frac{1}{2} + 3^0 + \frac{1}{2} = 2 = \frac{1}{2}(3^1 + 1);$
 $t_2 = 2 + 3^1 + \frac{1}{2} = 5\frac{1}{2} = \frac{1}{2}(3^2 + 2).$

Steps:

1. Prove S_a is true
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(b) Prove by mathematical induction that the statement holds true for all natural numbers n .

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Solution:

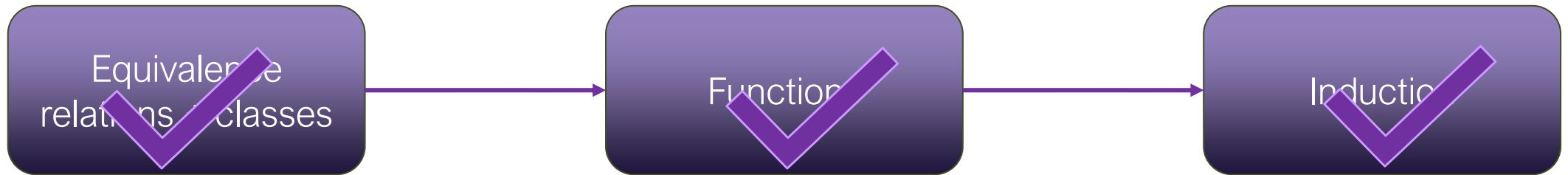
- *Base case:* $t_0 = \frac{1}{2} = \frac{1}{2}(3^0 + 0)$.
- *Induction step:* Let $m \geq 0$. Assume $t_m = \frac{1}{2}(3^m + m)$ (IH).

Then $t_{m+1} = t_m + 3^m + \frac{1}{2} \stackrel{\text{(IH)}}{=} \frac{1}{2}(3^m + m) + 3^m + \frac{1}{2} = \frac{3}{2} \cdot 3^m + \frac{1}{2}m + \frac{1}{2}$ and
 $\frac{1}{2}(3^{m+1} + (m+1)) = \frac{3}{2} \cdot 3^m + \frac{1}{2}m + \frac{1}{2}$.
Hence (IH) implies $t_{m+1} = \frac{1}{2}(3^{m+1} + (m+1))$.

Steps:

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Done!



Questions?

If you have any questions, ask them in the WhatsApp group!

Good Luck!
😊

Good luck with your final exam!

Practice Session Tomorrow !

Why choose the Exam Practice Session?

In this training, you will practice with exam level questions. Some of these questions tend to occur on the exam itself.

[Show less](#)

1 groups

Group 1

Last spots
Starts in 24 hours

Fr 22 Mar 09:00 - 12:00

€29,00 / €9,67 per hour

In cart 

Evaluation form



Step 1

Scan the QR code
or check your email



Step 2

Answer the 5
questions



Step 3

Sit back & relax

